

The maximal triangular Hilbert transform at the symmetric point

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Abstract

For arbitrary complex inputs in $L^3(\mathbb{R}^2)$, we prove the pointwise maximal $L^3 \times L^3 \rightarrow L^{3/2}$ estimate for the triangular Hilbert transform, with the supremum over both hard truncation endpoints. The estimate yields joint almost-everywhere and $L^{3/2}$ convergence as the lower endpoint tends to zero and the upper endpoint tends to infinity. This also proves the conjectured scalar estimate at the symmetric point.

Contents

1	Introduction	2
1.1	Previous work	2
1.2	Proof overview	3
2	A matrix energy and its Hessian	3
3	A mixed trace inequality	6
4	Gaussian heat dissipation	8
4.1	Differentiating the star energies	10
4.2	Comparing neighbors and integrating centers	11
4.3	The initial energy	12
5	Output-dependent truncations and energy jumps	13
5.1	Maximal averages on a line	13
5.2	Changing a matrix entry	15
5.3	The total jump bound	15
6	From masked energies to the maximal transform	18
6.1	Odd windows and selected endpoints	19
6.2	Finite maxima and the continuum limit	20
6.3	A pointwise comparison with hard truncations	22
7	General inputs and principal values	23
7.1	Scalar forms and the simplex coordinates	25

1 Introduction

The triangular Hilbert transform couples translations in two different coordinate directions. For complex functions F, G on $\mathbb{R}^2$ and $0 < \varepsilon < R < \infty$, write

$$B_{\varepsilon, R}(F, G)(x, y) = \int_{\varepsilon < |t| < R} F(x + t, y) G(x, y + t) \frac{dt}{t}, \quad B_*(F, G) = \sup_{0 < \varepsilon < R < \infty} |B_{\varepsilon, R}(F, G)|.$$

For L^3 inputs, the finite integrals are absolutely convergent outside one common null set, and the supremum is measurable; the details are given in Lemma 7.1.

Theorem 1.1. *There is an absolute constant C such that, for every pair of complex functions $F, G \in L^3(\mathbb{R}^2)$,*

$$\|B_*(F, G)\|_{L^{3/2}(\mathbb{R}^2)} \leq C \|F\|_{L^3(\mathbb{R}^2)} \|G\|_{L^3(\mathbb{R}^2)}.$$

The supremum is over both finite hard truncation endpoints.

The estimate controls a truncation chosen separately at each spatial point. This is stronger than a uniform bound for the norm of each individual truncation: the same $L^{3/2}$ majorant must control every choice of the two endpoints. It also gives the joint principal value

$$B(F, G) = \lim_{\substack{\varepsilon \downarrow 0 \\ R \uparrow \infty}} B_{\varepsilon, R}(F, G)$$

almost everywhere and in $L^{3/2}$, with the same norm bound (Theorem 7.2).

Pairing the bilinear output with a third L^3 function leads to the scalar triangular form. In its symmetric coordinates this is

$$\Lambda_{\varepsilon, R}(G_0, G_1, G_2) = \iiint_{\varepsilon < |a+b+c| < R} G_0(a, b) G_1(b, c) G_2(c, a) \frac{da \, db \, dc}{a + b + c}.$$

The three factors have equal exponent 3, which is the symmetric point of the scalar Hölder relation $1/p_0 + 1/p_1 + 1/p_2 = 1$. We obtain an endpoint-uniform bound by $C \prod_v \|G_v\|_3$ and a joint scalar principal value for every complex L^3 triple. Proposition 7.4 gives the explicit determinant-one changes of variables and proves that the flat and simplex scalar estimates have the same optimal constant. In particular, the result resolves positively the symmetric triangular Hilbert transform problem stated by Thiele in [8, Section 9, Problem 13].

1.1 Previous work

The flat operator appears in Demeter and Thiele's discussion of bilinear averages for two commuting transformations [3, Section 6]. The three inputs of the scalar form depend on the three pairs of variables, making it a basic example of an entangled singular integral. Kovač, Thiele and Zorin-Kranich proved estimates for a Walsh model with one structurally restricted input [13, Theorem 1.7].

Tao proved sublogarithmic cancellation for one-dimensional multilinear Hilbert transforms [17, Theorem 1.2]. Zorin-Kranich extended this cancellation to simplex Hilbert transforms [18, Theorem 1.3]. Square-function estimates of Durcik, Kovač, Škreb and Thiele gave square-root cancellation for the triangular form across finitely many smooth scales [5, Corollary 5]. Durcik, Kovač and Thiele proved power-type cancellation with hard truncations [6, Theorem 1]; at the symmetric tuple the bound grows like $\sqrt{\log(R/\varepsilon)}$, by permutation and interpolation of their $(4, 4, 2)$ estimates. These finite-scale bounds retain a dependence on the scale range.

Bounds for averages over direction parameters were obtained by Durcik and Roos [7, Theorem 1] and, for rough angular weights, by Lin and Slavíková [16, Theorem 1.1]. For curved translations, Christ, Durcik and Roos proved bounds along (t, t^2) [1, Theorem 1]. Hsu and Lin treat more general curves under geometric hypotheses excluding the flat curve (t, t) [10, Theorem 1.37]. The curvature in these results provides an additional source of cancellation that is unavailable for the flat curve.

Our argument uses a nonnegative matrix energy and Gaussian heat flow. Energy telescoping for entangled forms appears in Kovač's twisted-paraproduct argument [12, Lemma 2] and his Bellman-function framework [11, Section 2]. Continuous Gaussian telescoping appears in Durcik [4, Lemma 3 and Section 3]. These methods concern bipartite forms; Kovač discusses the obstacle posed by the triangular cycle in [11, Section 6.2]. The cyclic trace representation itself also occurs in [13, Equation (1.5)]. We prove a dimension-independent mixed trace inequality and compare the energy derivatives at neighboring vertices. The key additional estimate controls the total change in energy when individual matrix entries are switched on and off at independently chosen scales. This permits the pointwise maximum without a factor depending on the number of scales or the matrix dimensions.

1.2 Proof overview

The change of variables $x = -a$, $y = -b$, $t = a + b + c$ writes the bilinear output in terms of

$$G_1(b, c) = F(b + c, -b), \quad G_2(c, a) = G(-a, c + a).$$

Both pair maps preserve L^3 norms. We first sample these inputs, together with a dual input $G_0(a, b)$, on a finite mesh. Multiplying entries by Gaussian weights gives maps $W_v : H_{v+1} \rightarrow H_v$ between finite-dimensional complex Hilbert spaces, indexed by $v \in \mathbb{Z}/3\mathbb{Z}$. The three vertex spaces correspond to the sampled a -, b - and c -coordinates. Thus the two indices of the dual array record the output position $(x, y) = (-a, -b)$. At a vertex, combine its two incident maps into $R_v = [W_v \quad W_{v-1}^*]$ and use the energy $\text{tr}((R_v^* R_v)^{3/2})$.

Sections 2 and 3 prove the matrix inequalities. Section 4 then integrates the sum of the three energies over Gaussian centers satisfying $p_0 + p_1 + p_2 = 0$. For fixed arrays, its decrease as the variance grows controls the quadratic costs needed to estimate a cyclic trace with three odd Gaussian windows.

To select different truncations at different output points, the dual array entry at (i, j) must be present only between its own two scales. The energy is therefore decreasing between finitely many switching scales and may jump at those scales. Section 5 bounds the sum of the absolute jumps by the discrete L^3 masses of the three arrays. The underlying matrix fact is that the derivative with respect to an entry is controlled by the product of its row and column norms. After Gaussian integration, one-dimensional maximal functions bound these products uniformly in the switching scales.

Section 6 uses this bound to linearize any finite maximum of smooth truncations. Mesh limits and rational exhaustion give the full smooth maximum. A pointwise comparison of its kernel with the hard Hilbert kernel completes the estimate for compactly supported smooth inputs. Finally, Section 7 uses density and a vanishing tail-oscillation argument to prove Theorem 1.1, the joint principal values, and the scalar equivalence.

2 A matrix energy and its Hessian

We begin with the finite-dimensional inequalities that control the derivatives of our energy. Their constants are independent of the matrix dimensions. Throughout this section, matrices

act on finite-dimensional complex Hilbert spaces, and inequalities between Hermitian matrices mean inequalities of quadratic forms.

For a positive semidefinite matrix P , set

$$\Phi(P) = \text{tr}(P^{3/2}), \quad \text{supp } P = (\ker P)^\perp.$$

A matrix is *supported on* P if it vanishes on $\ker P$ on both sides. We use the Hilbert–Schmidt inner product $\langle K, L \rangle_{\text{HS}} = \text{tr}(K^*L)$. For a Hermitian matrix A , write $\mathsf{L}_A(X) = AX$ and $\mathsf{R}_A(X) = XA$. On matrices acting on $\text{supp } P$, the operator $\mathsf{L}_{\sqrt{P}} + \mathsf{R}_{\sqrt{P}}$ is positive definite. For Hermitian K, L supported on P , define

$$\mathcal{B}_P(K, L) = \frac{3}{2} \text{Re} \langle K, (\mathsf{L}_{\sqrt{P}} + \mathsf{R}_{\sqrt{P}})^{-1} L \rangle_{\text{HS}}, \quad b_P(K) = \mathcal{B}_P(K, K).$$

If $P = 0$, the only supported directions are zero, and these expressions are defined to be zero.

The square-root derivative used here is a case of the classical divided-difference calculus for Hermitian matrix functions [2, Section 1, Equation (1)]. We give a direct inverse-function proof to keep the support conventions explicit.

Lemma 2.1 (Energy derivatives). *On the cone of matrices positive definite on a fixed support, Φ is smooth. For Hermitian directions K, L on that support,*

$$d\Phi_P(K) = \frac{3}{2} \text{tr}(\sqrt{P}K), \quad d^2\Phi_P(K, L) = \mathcal{B}_P(K, L).$$

If $p_i > 0$ are the eigenvalues on the support, then in an orthonormal eigenbasis

$$b_P(K) = \frac{3}{2} \sum_{i,j} \frac{|K_{ij}|^2}{\sqrt{p_i} + \sqrt{p_j}}. \quad (2.1)$$

In particular, $\mathcal{B}_P$ is a positive definite symmetric bilinear form on the real space of supported Hermitian directions, and

$$|\mathcal{B}_P(K, L)| \leq b_P(K)^{1/2} b_P(L)^{1/2}.$$

Proof. Restrict to the support and put $A = \sqrt{P}$. The derivative of the squaring map at A is $X \mapsto AX + XA$. In an eigenbasis of A , this map multiplies its (i, j) entry by $\sqrt{p_i} + \sqrt{p_j} > 0$. The inverse function theorem therefore gives a smooth positive square root. Its derivative X_K in direction K satisfies

$$AX_K + X_KA = K.$$

Taking a trace after multiplication by A yields $\text{tr}(PX_K) = \frac{1}{2} \text{tr}(AK)$. Differentiation of $\Phi(P) = \text{tr}(P\sqrt{P})$ now gives

$$d\Phi_P(K) = \text{tr}(KA) + \text{tr}(PX_K) = \frac{3}{2} \text{tr}(AK).$$

Differentiating this identity in direction L proves the Hessian formula. The entrywise formula follows by diagonalizing A . It also proves symmetry, positivity, and Cauchy–Schwarz on the real space of Hermitian directions. $\square$

The energy and its Hessian are unchanged by simultaneous unitary conjugation of their arguments. They are also unchanged if all arguments are extended by zero to a larger Hilbert space: the positive eigenvalues and the supported entries in (2.1) stay the same. We will use these observations to compare Grams acting on different subspaces.

Lemma 2.2 (Order comparison). *Let P, Q be positive semidefinite matrices on the same Hilbert space, let $a \geq 1$, and suppose $P \leq aQ$. Every Hermitian K supported on P is supported on Q , and*

$$b_P(K) \geq a^{-1/2} b_Q(K). \quad (2.2)$$

Proof. If $x \in \ker Q$, then $0 \leq \langle Px, x \rangle \leq a \langle Qx, x \rangle = 0$, so $Px = 0$. Thus $\ker Q \subseteq \ker P$, proving the support assertion.

We recall the two operator-order facts needed for the comparison. Inversion reverses order on positive definite matrices: for $0 < X \leq Y$, conjugate by $X^{-1/2}$, invert the resulting matrix at least I , and conjugate back. Moreover,

$$\sqrt{X} = \frac{1}{\pi} \int_0^\infty X(X+tI)^{-1} \frac{dt}{\sqrt{t}}.$$

This norm-convergent formula follows from the scalar integral by spectral calculus. Since $X(X+tI)^{-1} = I - t(X+tI)^{-1}$, inversion order proves that the positive square root preserves order. This is the classical Loewner–Heinz square-root monotonicity; see, for example, Kwong [14, Theorem 1 and Section 2]. The supported Hessian comparison below is derived from these order facts here.

For $\eta > 0$, the assumptions give

$$P + \eta I \leq a(Q + \eta I), \quad \sqrt{P + \eta I} \leq \sqrt{a} \sqrt{Q + \eta I}.$$

If $A \leq B$, then $L_A + R_A \leq L_B + R_B$ as selfadjoint operators on Hilbert–Schmidt space. Indeed, left and right multiplication by the positive matrix $B - A$ have the nonnegative quadratic forms $\text{tr}(X^*(B - A)X)$ and $\text{tr}(X^*X(B - A))$. Inverting the resulting positive definite operators gives

$$b_{P+\eta I}(K) \geq a^{-1/2} b_{Q+\eta I}(K).$$

Finally let $\eta \downarrow 0$. Formula (2.1), applied in a full eigenbasis, shows that both limits are the asserted supported costs: every entry of K meeting the kernel of the respective matrix is zero. $\square$

We will apply the Hessian to Gram matrices with an insertion between the two factors. The following bound controls such derivatives by the energy itself.

Lemma 2.3 (Inserted Gram bound). *Let $R : E \rightarrow F$ be a linear map between finite-dimensional complex Hilbert spaces, and let $L : F \rightarrow F$ be Hermitian. The matrix R^*LR is supported on R^*R , and*

$$b_{R^*R}(R^*LR) \leq \frac{3}{4} \|L\|_{\text{op}}^2 \Phi(R^*R). \quad (2.3)$$

Proof. The insertion annihilates $\ker R = \ker(R^*R)$ and is Hermitian, so it is supported on the Gram. The assertion is immediate if $R = 0$. Otherwise take a singular-value decomposition, with positive singular values ρ_i , and let C be the compression of L to the corresponding left singular vectors. Then $\|C\|_{\text{op}} \leq \|L\|_{\text{op}}$ and

$$b_{R^*R}(R^*LR) = \frac{3}{2} \sum_{i,j} \frac{\rho_i^2 \rho_j^2}{\rho_i + \rho_j} |C_{ij}|^2.$$

For $x, y > 0$, the arithmetic–geometric mean inequality gives

$$\frac{x^2 y^2}{x + y} \leq \frac{(xy)^{3/2}}{2} \leq \frac{x^3 + y^3}{4}.$$

Each row and each column of C has squared Euclidean norm at most $\|L\|_{\text{op}}^2$. Consequently

$$b_{R^*R}(R^*LR) \leq \frac{3}{8} \sum_{i,j} (\rho_i^3 + \rho_j^3) |C_{ij}|^2 \leq \frac{3}{4} \|L\|_{\text{op}}^2 \sum_i \rho_i^3,$$

which is (2.3). $\square$

3 A mixed trace inequality

The triangular form is naturally represented by a cyclic product of operators. Such a representation also appears in the dyadic model of Kovač, Thiele and Zorin-Kranich [13, Equation (1.5)]. We prove a finite-dimensional estimate that controls this product by three of the quadratic costs from Section 2.

Let H_v , $v \in \mathbb{Z}/3\mathbb{Z}$, be finite-dimensional complex Hilbert spaces, and let $W_v : H_{v+1} \rightarrow H_v$ be arbitrary linear maps. Combine the two edges incident to v into the *row star*

$$R_v = \begin{bmatrix} W_v & W_{v-1}^* \end{bmatrix} : H_{v+1} \oplus H_{v-1} \longrightarrow H_v. \quad (3.1)$$

Its two Gram matrices are

$$T_v = R_v^* R_v, \quad S_v = R_v R_v^*. \quad (3.2)$$

Their positive eigenvalues are the squares of the nonzero singular values of R_v , with the same multiplicities. In particular, $\Phi(T_v) = \Phi(S_v)$. Figure 1 shows the cyclic edges and the row star at vertex 0.

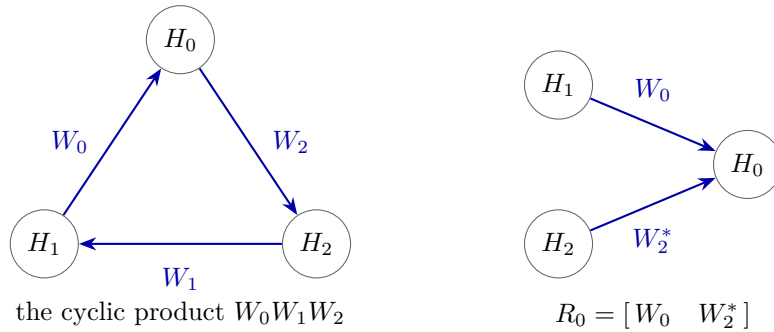


Figure 1: The three edge maps and the row star at vertex 0. Taking the adjoint of W_2 makes both blocks of R_0 have target H_0 . The two Grams $R_0^* R_0$ and $R_0 R_0^*$ have the same positive eigenvalues on different spaces.

Lemma 3.1 (Mixed trace estimate). *Let $W_v : H_{v+1} \rightarrow H_v$ be linear maps between finite-dimensional complex Hilbert spaces, with row stars R_v and Gram matrices T_v as in (3.1)–(3.2). For Hermitian maps $D_v : H_v \rightarrow H_v$, put $D_* = \max_v \|D_v\|_{\text{op}}$. Then*

$$|\text{tr}(D_0 W_0 D_1 W_1 D_2 W_2)| \leq \frac{5}{3} D_* \sum_{v=0}^2 b_{T_v}(R_v^* D_v R_v). \quad (3.3)$$

The constant is independent of the dimensions of the spaces.

Proof. We first put the three quadratic costs in a common metric. We then estimate a cubic trace in an eigenbasis and recover the absolute value of the original cyclic product by changing the phase of one edge.

A common quadratic form. On $H = H_0 \oplus H_1 \oplus H_2$, define the Hermitian matrices

$$M = \begin{pmatrix} 0 & W_0 & W_2^* \\ W_0^* & 0 & W_1 \\ W_2 & W_1^* & 0 \end{pmatrix}, \quad D = \text{diag}(D_0, D_1, D_2).$$

Thus $\|D\|_{\text{op}} = D_*$. Let P_v be the orthogonal projection onto H_v and set

$$A_v = MP_vM, \quad K_v = MP_vDM.$$

Only the v -th block row of P_vM can be nonzero; it consists of R_v with its columns placed in the corresponding vertex spaces. Consequently A_v and K_v are, respectively, T_v and $R_v^*D_vR_v$, after a permutation of blocks and extension by zero. These operations preserve the quadratic cost. Since P_v commutes with D , the matrix K_v is Hermitian. Moreover,

$$\ker A_v = \ker(P_vM) \subseteq \ker K_v, \quad 0 \leq A_v \leq M^2.$$

The first identity follows from $\langle A_v x, x \rangle = \|P_v M x\|^2$. Thus K_v is supported on A_v , and hence also on M^2 . Lemma 2.2, followed by the quadratic-form inequality for three vectors, yields

$$\begin{aligned} \mathcal{E} &:= \sum_v b_{T_v}(R_v^* D_v R_v) \\ &\geq \sum_v b_{M^2}(K_v) \geq \frac{1}{3} b_{M^2} \left(\sum_v K_v \right) = \frac{1}{3} b_{M^2}(MDM). \end{aligned} \tag{3.4}$$

All these quadratic forms are taken on their supports; no invertibility of M is required.

The cubic trace. Choose an orthonormal eigenbasis (e_i) of M , write the real eigenvalues as m_i , and arrange the indices so that $q_i := |m_i|$ is nonincreasing. Equal values may be ordered arbitrarily. Write D_{ij} for the entries of D in this basis. The spectral formula for b_{M^2} gives

$$\mathcal{E} \geq \frac{1}{2} Q, \quad Q := \sum_{i,j: q_i + q_j > 0} \frac{q_i^2 q_j^2}{q_i + q_j} |D_{ij}|^2. \tag{3.5}$$

If either eigenvalue is zero, its summand is zero; pairs of zero eigenvalues are omitted. Define

$$A_i = q_i^3 |D_{ii}|^2, \quad E_i = q_i \sum_{j>i} q_j^2 |D_{ij}|^2, \quad A = \sum_i A_i, \quad E = \sum_i E_i.$$

Hermitian symmetry and the ordering of the q_i imply

$$Q = \frac{A}{2} + 2 \sum_{\substack{i<j \\ q_i + q_j > 0}} \frac{q_i^2 q_j^2}{q_i + q_j} |D_{ij}|^2 \geq \frac{A}{2} + E. \tag{3.6}$$

Indeed, for $i < j$ we have $q_j \leq q_i$, so $2q_i/(q_i + q_j) \geq 1$ whenever the denominator is positive.

We now bound

$$\text{tr}((DM)^3) = \sum_{i,j,k} m_i m_j m_k D_{ij} D_{jk} D_{ki}$$

by grouping the triples according to how often their smallest index occurs. This is the index of a largest absolute eigenvalue in the triple.

If the smallest index i occurs once, its three cyclic positions give

$$3m_i \sum_{j,k>i} m_j m_k D_{ij} D_{jk} D_{ki}.$$

Here $j = k$ is allowed. For the vector $x = (m_j D_{ji})_{j>i}$, the inner sum is $x^* D_{>i} x$, where $D_{>i}$ is the compression of D to the indices greater than i . Since $\|D_{>i}\|_{\text{op}} \leq D_*$, the modulus of this contribution is at most $3D_* E_i$. This argument retains the signs of the m_j and uses $D_{ji} = \overline{D_{ij}}$.

If the smallest index i occurs twice, the contribution is

$$3m_i^2 D_{ii} \sum_{k>i} m_k |D_{ik}|^2.$$

Cauchy–Schwarz and $\sum_k |D_{ik}|^2 = \|De_i\|^2 \leq D_*^2$ give

$$\left| \sum_{k>i} m_k |D_{ik}|^2 \right| \leq D_* \left(\sum_{k>i} q_k^2 |D_{ik}|^2 \right)^{1/2}.$$

Its contribution therefore has modulus at most $3D_* \sqrt{A_i E_i}$. This also holds when $q_i = 0$, without any division by q_i .

Finally, when all three indices equal i , the modulus is at most $D_* A_i$. These three cases exhaust all triples, including repeated indices. Summing them and applying Cauchy–Schwarz once more, we obtain

$$\begin{aligned} |\text{tr}((DM)^3)| &\leq D_*(A + 3E + 3\sqrt{AE}) \\ &\leq D_* \left(\frac{5}{2}A + \frac{9}{2}E \right) \leq 5D_* \left(\frac{A}{2} + E \right) \leq 5D_* Q. \end{aligned} \tag{3.7}$$

Recovering the complex cyclic product. We have bounded the cubic trace by the star costs. It remains to extract the modulus of the desired triangular trace. Put

$$\tau = \text{tr}(D_0 W_0 D_1 W_1 D_2 W_2).$$

Because M has zero diagonal blocks, a nonzero closed three-step block product must visit all three vertices. The three starting points in one orientation each give τ , by cyclicity of the trace. Reversing the orientation gives $\bar{\tau}$, since the D_v are Hermitian. Hence

$$\text{tr}((DM)^3) = 6 \text{Re } \tau.$$

If $\tau \neq 0$, choose $z \in \mathbb{C}$ with $|z| = 1$ and $z\tau = |\tau|$, and replace W_0 by zW_0 . The row stars change by

$$\begin{aligned} R_0 &\mapsto R_0 \text{diag}(zI_{H_1}, I_{H_2}), \\ R_1 &\mapsto R_1 \text{diag}(I_{H_2}, \bar{z}I_{H_0}), \end{aligned}$$

while R_2 is unchanged. Each displayed right multiplier is unitary and conjugates the corresponding Gram and inserted Gram by the same unitary. Thus $\mathcal{E}$ is unchanged, as is D_* . Applying (3.5) and (3.7) to the modified matrices, with Q computed for the new M , gives

$$6|\tau| \leq 5D_* Q \leq 10D_* \mathcal{E},$$

which is (3.3). The case $\tau = 0$ is immediate. $\square$

4 Gaussian heat dissipation

We now weight the three edge matrices by Gaussians. For the variance rates chosen below, the sum of the star energies decreases after integration over a plane of centers. This decrease

controls the matrix derivatives that will produce the odd kernel. The estimate must be independent of the number of matrix entries; this uniformity will allow us to pass from finite matrices to functions.

Continuous Gaussian telescoping for entangled forms appears in Durcik [4, Lemma 3 and Section 3]. Here we use the Hessian of a matrix energy and compare its costs at neighboring vertices.

Fix $h > 0$, an integer $N \geq 0$, and nodes $u_i = ih$, $-N \leq i \leq N$. At vertex $v \in \mathbb{Z}/3\mathbb{Z}$ take $H_v = \mathbb{C}^{2N+1}$, and prescribe arbitrary complex edge values $A_v(i, j)$. For centers $p_v \in \mathbb{R}$ and variances $\sigma_v > 0$, define

$$g_\sigma(t) = (2\pi\sigma)^{-1/2} e^{-t^2/(2\sigma)}, \quad w_v(i) = h g_{\sigma_v}(u_i - p_v),$$

$$W_v(i, j) = A_v(i, j) \sqrt{w_v(i) w_{v+1}(j)}.$$

Use the stars $R_v = [W_v \ W_{v-1}^*]$ and Grams $T_v = R_v^* R_v$, $S_v = R_v R_v^*$ from Section 3, and set

$$e_v = \Phi(T_v) = \Phi(S_v).$$

Define the diagonal score and its insertions into the incident stars by

$$L_w = \text{diag}_i \frac{u_i - p_w}{\sigma_w}, \quad U_v = R_v^* L_v R_v,$$

$$U_{v,v+1} = W_v L_{v+1} W_v^*, \quad U_{v,v-1} = W_{v-1}^* L_{v-1} W_{v-1}.$$

Define their quadratic costs and the mixed term by

$$V_v = b_{T_v}(U_v), \quad Y_{v,w} = b_{S_v}(U_{v,w}) \quad (w \neq v), \quad Z_v = \mathcal{B}_{S_v}(U_{v,v+1}, U_{v,v-1}). \quad (4.1)$$

Every insertion is supported on the Gram used in its definition: an inserted Gram $A^* B A$ annihilates $\ker A$ on both sides.

For $d \in \mathbb{R}$, let $\Pi_d = \{p_0 + p_1 + p_2 = d\}$, with the measure

$$\int_{\Pi_d} H \, d\pi_d = \int_{\mathbb{R}^2} H(p_0, p_1, d - p_0 - p_1) \, dp_0 \, dp_1. \quad (4.2)$$

Any other pair of free coordinates gives the same measure, since the change of coordinates has absolute determinant one. Put $\lambda = 11/10$.

We also write

$$\mathcal{S}(A) = \sum_{v=0}^2 \sum_{i,j=-N}^N h^2 |A_v(i, j)|^3.$$

Our goal in this section is the fixed-array inequality

$$-\partial_s J(d, s) \geq c_* \int_{\Pi_d} \sum_v V_v \, d\pi_d, \quad J(d, s) = \int_{\Pi_d} \sum_v e_v \, d\pi_d,$$

when the variances are $\sigma_i = \alpha_i s$ and α is a permutation of $(1, \lambda, \lambda)$. The odd-window estimate in Section 6 uses wider Gaussian weights for uniformly bounded diagonal factors, but compares each score cost after narrowing only its own vertex. The fixed enlargement $\lambda = 11/10$ retains strict dissipation for all three comparisons. Lemma 4.4 gives an absolute $c_* > 0$; Lemma 4.5 bounds J by $C\mathcal{S}(A)$ at scales $s \geq h^2$. Later, when entries change at prescribed scales, the same differential inequality will apply between successive changes.

We first differentiate the star energies, choosing the Gram orientation whose support stays fixed. We then integrate over the center plane and compare neighboring stars. These two steps produce the strict decrease that controls the score costs.

4.1 Differentiating the star energies

Lemma 4.1. *For fixed finite edge arrays, each e_v is smooth in all centers and positive variances. If only the parameters at vertex v vary, the support of T_v is fixed. If the parameters at either or both other vertices vary while those at v are fixed, the support of S_v is fixed.*

Proof. The weighted star has the form $R_v = D_{\text{row}} R_v^0 D_{\text{col}}$, where $R_v^0 = [A_v \ A_{v-1}^*]$ is fixed and both diagonal multipliers are strictly positive; the column multiplier has one block for each neighbor. Invertible row multiplication preserves the kernel, and invertible column multiplication preserves the range. This proves the two support assertions, since

$$\ker R_v = \ker(R_v^0 D_{\text{col}}), \quad \text{ran } R_v = D_{\text{row}} \text{ran } R_v^0$$

when the respective other multiplier is held fixed.

For joint smoothness, the rank-zero case is immediate. Otherwise factor $R_v^0 = XY^*$ with X, Y of full column rank. Write $X' = D_{\text{row}} X$, $Y' = D_{\text{col}} Y$ and $P' = X'^* X'$, $Q' = Y'^* Y'$. These last two matrices are positive definite and depend smoothly on the parameters. The nonzero eigenvalues of $R_v R_v^* = X' Q' X'^*$ are those of $Q'^{1/2} P' Q'^{1/2}$, by comparing AA^* and $A^* A$ for $A = X' Q'^{1/2}$. Therefore

$$e_v = \Phi(Q'^{1/2} P' Q'^{1/2}),$$

which is smooth on the positive definite cone. $\square$

Lemma 4.2. *Fix a positive vector α and set $\sigma_i = \alpha_i s$. For all centers and all $s > 0$,*

$$2\partial_s e_v = \sum_i \alpha_i \partial_{p_i}^2 e_v - \alpha_v V_v - \sum_{w \neq v} \alpha_w Y_{v,w}, \quad (4.3)$$

$$\partial_{p_{v+1}} \partial_{p_{v-1}} e_v = Z_v. \quad (4.4)$$

Proof. For $w = hg_\sigma(u - p)$ and $L = (u - p)/\sigma$,

$$\partial_p w = Lw, \quad \partial_p^2 w = (L^2 - \sigma^{-1})w, \quad \partial_\sigma w = \frac{1}{2} \partial_p^2 w.$$

With the columns fixed, T_v is linear in the row density, so

$$\partial_{p_v} T_v = U_v, \quad \partial_{\sigma_v} T_v = \frac{1}{2} \partial_{p_v}^2 T_v.$$

Its support is fixed by Lemma 4.1. The chain rule and Lemma 2.1 thus give

$$\partial_{\sigma_v} e_v = \frac{1}{2} \partial_{p_v}^2 e_v - \frac{1}{2} \mathcal{B}_{T_v}(U_v, U_v) = \frac{1}{2} \partial_{p_v}^2 e_v - \frac{1}{2} V_v.$$

For a column vertex $w \neq v$, use instead S_v , which is linear in that column density and has fixed support. This gives

$$\partial_{p_w} S_v = U_{v,w}, \quad \partial_{\sigma_w} e_v = \frac{1}{2} \partial_{p_w}^2 e_v - \frac{1}{2} Y_{v,w}.$$

Summing the three variance derivatives with coefficients α_i proves (4.3).

The two summands in $S_v = W_v W_v^* + W_{v-1}^* W_{v-1}$ depend on different column densities. Its mixed derivative in the two column centers is therefore zero. Only the Hessian term remains in the mixed derivative of e_v , giving (4.4). $\square$

4.2 Comparing neighbors and integrating centers

Lemma 4.3. *For every choice of the weighted matrices and every vertex w ,*

$$Y_{w+1,w} + Y_{w-1,w} \leq \sqrt{2} V_w. \quad (4.5)$$

Proof. Abbreviate $A = W_w$, $B = W_{w-1}^*$, and $L = L_w$. The inequality $\|Az + Bz'\|^2 \leq 2\|Az\|^2 + 2\|Bz'\|^2$ implies

$$T_w \leq 2Q, \quad Q = \text{diag}(A^*A, B^*B).$$

Since $U_w = [A \ B]^* L [A \ B]$ is supported on T_w , Lemma 2.2 gives $b_{T_w}(U_w) \geq 2^{-1/2} b_Q(U_w)$. In a block-preserving eigenbasis of Q , the formula for b_Q is a sum of nonnegative squared entries. Discarding the off-diagonal blocks yields

$$b_Q(U_w) \geq b_{A^*A}(A^*LA) + b_{B^*B}(B^*LB).$$

Here $A^*A \leq S_{w+1}$ and $B^*B \leq S_{w-1}$, and the respective insertions are $U_{w+1,w}$ and $U_{w-1,w}$. Each is supported on its smaller edge Gram. Apply Lemma 2.2 again, now with comparison factor one, to obtain (4.5). $\square$

Lemma 4.4. *Fix finite edge arrays and a positive vector α , and evaluate the energies at $\sigma_i = \alpha_i s$. Set*

$$J_v(d, s) = \int_{\Pi_d} e_v \, d\pi_d, \quad J(d, s) = \sum_v J_v(d, s), \quad \Sigma_\alpha = \sum_i \alpha_i.$$

These integrals are finite, continuously differentiable in $s > 0$, and twice continuously differentiable in d . Moreover,

$$2\partial_s J(d, s) \leq \int_{\Pi_d} \sum_w \left[-\alpha_w + \sqrt{2} \max\{0, \Sigma_\alpha/2 - \alpha_w\} \right] V_w \, d\pi_d. \quad (4.6)$$

In particular, for each permutation of $(1, \lambda, \lambda)$,

$$-\partial_s J(d, s) \geq c_* \int_{\Pi_d} \sum_w V_w \, d\pi_d, \quad c_* = \frac{1}{2} \min\{1 - \frac{3}{5}\sqrt{2}, \frac{11}{10} - \frac{1}{2}\sqrt{2}\} > 0. \quad (4.7)$$

Proof. We first justify differentiation at the fixed mesh. On compact positive s -intervals the nodes are bounded, the scores grow at most linearly in the centers, and the first and second density ratios grow at most quadratically. Lemma 2.3 gives

$$V_v \leq \frac{3}{4} \|L_v\|_{\text{op}}^2 e_v, \quad Y_{v,w} \leq \frac{3}{4} \|L_w\|_{\text{op}}^2 e_v.$$

For the second inequality, pad L_w by zero on the other column block, obtaining B_w with $U_{v,w} = R_v B_w R_v^*$, and apply the lemma to R_v^* . Also $2|Z_v| \leq Y_{v,v+1} + Y_{v,v-1}$. For a Hermitian matrix B ,

$$|\text{tr}(\sqrt{T_v} R_v^* B R_v)| \leq \|B\|_{\text{op}} e_v,$$

since $R_v \sqrt{T_v} R_v^*$ is positive with trace e_v . The corresponding column estimate follows in the same way with S_v and a padded column insertion. Together with the density-ratio formulas in Lemma 4.2, these bounds control every first center derivative, pure second center derivative, mixed column derivative, and first s -derivative by a polynomial in the centers times e_v .

To dominate the energy itself, use

$$e_v \leq (\|W_v\|_{\text{HS}}^2 + \|W_{v-1}\|_{\text{HS}}^2)^{3/2} \leq \sqrt{2} (\|W_v\|_{\text{HS}}^3 + \|W_{v-1}\|_{\text{HS}}^3). \quad (4.8)$$

Each edge norm cubed is bounded by a Gaussian in its two endpoint centers, locally uniformly for $s > 0$. Indeed, the finite set of nodes is bounded and $(u - p)^2 \geq p^2/2 - u^2$. On Π_d with $|d| \leq D$, any pair of centers satisfies

$$p_0^2 + p_1^2 + p_2^2 \leq 4(p_i^2 + p_j^2) + 3D^2.$$

Thus an endpoint-pair Gaussian, multiplied by any of the preceding polynomials, has an integrable majorant in any fixed plane chart, locally uniformly in (d, s) . This proves all needed differentiation statements by dominated convergence. The constants in this justification may depend on the finite mesh and the compact parameter intervals; they will not enter the uniform estimate.

We can now differentiate the plane integrals. It remains to express their second derivatives in d using both pure and mixed center derivatives, then apply Lemma 4.3 to the resulting costs.

Choosing p_i as the dependent coordinate in (4.2) gives

$$\partial_d^2 J_v = \int_{\Pi_d} \partial_{p_i}^2 e_v \, d\pi_d \quad \text{for every } i.$$

For the column indices $j = v + 1$, $k = v - 1$, first choose p_j dependent to write $\partial_d J_v = \int_{\Pi_d} \partial_{p_j} e_v \, d\pi_d$. Reparametrize this last integral with p_k dependent, then differentiate again. It follows from (4.4) that

$$\partial_d^2 J_v = \int_{\Pi_d} \partial_{p_k} \partial_{p_j} e_v \, d\pi_d = \int_{\Pi_d} Z_v \, d\pi_d. \quad (4.9)$$

These are identities of integrated derivatives; no pointwise equality between pure and mixed derivatives is required.

Integrating (4.3) and using (4.9), we obtain

$$2\partial_s J_v = \int_{\Pi_d} \left[\Sigma_\alpha Z_v - \alpha_v V_v - \sum_{w \neq v} \alpha_w Y_{v,w} \right] d\pi_d.$$

Cauchy–Schwarz gives $2Z_v \leq Y_{v,v+1} + Y_{v,v-1}$. Sum over v and collect the two neighbor costs carrying the score at w . Their common coefficient is $\Sigma_\alpha/2 - \alpha_w$. If it is negative, discard those terms; otherwise apply Lemma 4.3. This proves (4.6).

For the indicated permutations, $\Sigma_\alpha = 16/5$. The two possible coefficients in (4.6) are $-1 + (3/5)\sqrt{2}$ and $-11/10 + (1/2)\sqrt{2}$, both strictly negative. Dividing by two gives (4.7). $\square$

4.3 The initial energy

The previous lemma controls the accumulated costs by the energy lost along the heat flow. We now bound the energy itself at scales no smaller than the mesh. This estimate will also apply immediately after entries are switched on or off, since it requires only a bound on their absolute values.

Lemma 4.5 (Initial energy). *Let $h > 0$, let $N \geq 0$ be an integer, and let $A_v(i, j)$ be the finite complex arrays above. For any $d \in \mathbb{R}$ and variances $\sigma_v \geq h^2$,*

$$\int_{\Pi_d} \sum_v e_v \, d\pi_d \leq C_0 \mathcal{S}(A), \quad C_0 = 2\sqrt{2} \left(1 + \sqrt{2/\pi} \right). \quad (4.10)$$

The same bound holds if any entries are replaced by smaller entries in absolute value before forming the weighted matrices.

Proof. For every center p and variance $\sigma > 0$,

$$\sum_{i=-N}^N h g_\sigma(u_i - p) \leq 1 + \frac{2h}{\sqrt{2\pi\sigma}}. \quad (4.11)$$

To see this, compare a grid point $u_i \geq p + h$ with the integral over its preceding interval of length h , where the Gaussian is at least its value at u_i . For $u_i \leq p - h$, use the succeeding interval. These intervals are disjoint, and the at most two remaining grid points each contribute at most $h \sup g_\sigma$.

Since each $\sigma_v \geq h^2$, every density has total mass at most $M = 1 + \sqrt{2/\pi}$. Weighted Hölder gives

$$\begin{aligned} \|W_v\|_{\text{HS}}^3 &= \left(\sum_{i,j} |A_v(i,j)|^2 w_v(i) w_{v+1}(j) \right)^{3/2} \\ &\leq M \sum_{i,j} |A_v(i,j)|^3 w_v(i) w_{v+1}(j). \end{aligned}$$

Integrate with p_v, p_{v+1} as free coordinates on Π_d . Each density has integral h , so

$$\int_{\Pi_d} \|W_v\|_{\text{HS}}^3 d\pi_d \leq M \sum_{i,j} h^2 |A_v(i,j)|^3.$$

Each edge occurs in exactly two stars. Equation (4.8) therefore proves (4.10). Replacing entries by smaller ones in absolute value only reduces the last upper bounds, proving the final assertion. $\square$

For fixed arrays and a permutation α of $(1, \lambda, \lambda)$, Lemmas 4.4 and 4.5 now give, whenever $h^2 \leq s_0 < s_1$,

$$\int_{s_0}^{s_1} \int_{\Pi_0} \sum_v V_v d\pi_0 ds \leq \frac{J(0, s_0) - J(0, s_1)}{c_*} \leq \frac{C_0}{c_*} \mathcal{S}(A).$$

The first inequality retains the final energy. In the next section, we will add these inequalities over intervals with different masked arrays.

5 Output-dependent truncations and energy jumps

We now allow each entry of one edge array to have its own truncation interval. Between endpoint scales, the heat estimate from Section 4 still applies. The new task is to bound the total increase in energy when entries switch on or off. We will in fact bound the sum of the absolute energy jumps. The estimate depends on the original arrays alone, because every entry switches at most twice. We first establish two inputs to this bound: estimates for Gaussian averages that are uniform in the scale, and a formula for changes in the matrix energy that remains valid when the rank changes.

5.1 Maximal averages on a line

For a locally integrable function on $\mathbb{R}$, and a sequence on $\mathbb{Z}$, respectively, define the centered maximal functions

$$M_{\mathbb{R}} f(x) = \sup_{r>0} \frac{1}{2r} \int_{x-r}^{x+r} |f(t)| dt, \quad M_{\mathbb{Z}} f(n) = \sup_{r \in \mathbb{Z}_{\geq 0}} \frac{1}{2r+1} \sum_{|m-n| \leq r} |f(m)|.$$

Thus discrete averages include singletons. We use Lebesgue measure on $\mathbb{R}$ and counting measure on $\mathbb{Z}$. The estimates below are versions of the classical Hardy–Littlewood maximal inequalities [9, Section IV]; we include the centered covering argument and the Gaussian comparison with the constants needed here.

Lemma 5.1 (One-dimensional maximal estimates). *On either of these measure spaces, write M for the corresponding centered maximal operator. For $f \in L^1$ and $\rho > 0$,*

$$\mu\{Mf > \rho\} \leq \frac{3}{\rho} \|f\|_1.$$

For every $1 < q < \infty$ and $f \in L^q$,

$$\|Mf\|_q^q \leq \frac{6q 2^{q-1}}{q-1} \|f\|_q^q. \quad (5.1)$$

Moreover, there is an absolute constant C_g such that, whenever $h > 0$, $\sigma \geq h^2$ and $n \in \mathbb{Z}$,

$$\sum_{m \in \mathbb{Z}} hg_\sigma((m-n)h) |f(m)| \leq C_g M_{\mathbb{Z}} f(n). \quad (5.2)$$

Proof. For a finite collection of intervals, repeatedly choose an interval of largest radius and discard all intervals meeting it. The chosen intervals are disjoint. Every discarded interval has radius no larger than that of the interval which removed it and is contained in its triple-radius enlargement. This enlargement has three times the original length on $\mathbb{R}$. On $\mathbb{Z}$, its cardinality is $6r+1 \leq 3(2r+1)$, including $r=0$.

For each point in a finite subset of $\{M_{\mathbb{Z}}f > \rho\}$, choose a witnessing interval whose average exceeds ρ . The preceding selection bounds the size of that subset by three times the sum of the cardinalities of the chosen intervals, and hence by $3\|f\|_1/\rho$. Taking the supremum over finite subsets proves the discrete weak estimate. On $\mathbb{R}$, fixed-radius averages of an L^1 function are continuous, so $\{M_{\mathbb{R}}f > \rho\}$ is open. Apply the same selection to a finite witnessing cover of any compact subset and then use inner regularity to obtain the continuous estimate.

For $f \in L^q$, split f into its parts on $\{|f| \leq \rho/2\}$ and $\{|f| > \rho/2\}$. The first part has maximal function at most $\rho/2$, and the second is integrable. The weak estimate gives

$$\mu\{Mf > \rho\} \leq \frac{6}{\rho} \int_{|f| > \rho/2} |f| \, d\mu.$$

The distribution formula and Tonelli's theorem now yield

$$\begin{aligned} \|Mf\|_q^q &\leq 6q \int |f(x)| \int_0^{2|f(x)|} \rho^{q-2} \, d\rho \, d\mu(x) \\ &= \frac{6q 2^{q-1}}{q-1} \|f\|_q^q. \end{aligned}$$

These formulas are valid first with a bounded range of distribution parameters and then by monotone convergence; finiteness of Mf is not assumed in advance. Both maximal functions are measurable: the discrete supremum is countable, and continuous averages are continuous in their radius, so positive rational radii suffice on $\mathbb{R}$.

Finally put $a = \sqrt{\sigma}/h \geq 1$. In the sum in (5.2), the kernel is

$$hg_\sigma((m-n)h) = \frac{1}{\sqrt{2\pi}a} \exp\left(-\frac{(m-n)^2}{2a^2}\right).$$

Split the integers into $|m - n| \leq a$ and the annuli $2^{j-1}a < |m - n| \leq 2^j a$, $j \geq 1$. The sum of $|f|$ over $|m - n| \leq 2^j a$ is at most $3 \cdot 2^j a M_{\mathbb{Z}} f(n)$. Bounding the Gaussian by its maximum on each region proves the claim with, for example,

$$C_g = \frac{3}{\sqrt{2\pi}} \left(1 + \sum_{j \geq 1} 2^j e^{-4^{j-1}/2} \right) < \infty.$$

All constants are independent of the mesh and of the support of f . $\square$

5.2 Changing a matrix entry

At a truncation endpoint the rank of a star can change. We therefore need a first-derivative formula that does not require fixed support. For the sum of cubes of the singular values, this formula is a case of the general gradient formula for unitarily invariant matrix functions [15, Theorem 3.1]. We give a direct regularization proof and then establish the entrywise bound used in the jump estimate.

Lemma 5.2 (A derivative through rank changes). *For finite rectangular complex matrices R_-, R_+ of the same size, put $R(z) = (1 - z)R_- + zR_+$ and $D = R_+ - R_-$. Then*

$$\begin{aligned} & \Phi(R_+^* R_+) - \Phi(R_-^* R_-) \\ &= 3 \int_0^1 \operatorname{Re} \langle R(z) \sqrt{R(z)^* R(z)}, D \rangle_{\text{HS}} \, dz. \end{aligned} \quad (5.3)$$

For every rectangular matrix R and every entry (i, j) ,

$$|(R \sqrt{R^* R})_{ij}| \leq \|\operatorname{row}_i R\|_2 \|\operatorname{col}_j R\|_2. \quad (5.4)$$

Proof. For $\eta > 0$, differentiate $\operatorname{tr}((R(z)^* R(z) + \eta I)^{3/2})$ using Lemma 2.1. Its derivative is

$$3 \operatorname{Re} \langle R(z) \sqrt{R(z)^* R(z) + \eta I}, D \rangle_{\text{HS}}.$$

Spectral calculus gives

$$\|\sqrt{R^* R + \eta I} - \sqrt{R^* R}\|_{\text{op}} \leq \sqrt{\eta}.$$

Since $R(z)$ stays in a bounded set, the derivatives converge uniformly in z , and the endpoint energies converge. Integrating the regularized identity before letting $\eta \downarrow 0$ proves (5.3).

Cauchy–Schwarz in the matrix-product formula bounds the left side of (5.4) by the norm of row i of R times the norm of column j of $\sqrt{R^* R}$. The squared norm of that column is $(R^* R)_{jj}$, which is also the squared norm of column j of R . This proves (5.4), including zero columns and singular matrices. $\square$

5.3 The total jump bound

Return to the finite arrays and Gaussian weights of Section 4, and write $d\pi = d\pi_0$ on Π_0 . Independently for every pair (i, j) , choose finite endpoints

$$h \leq \varepsilon_{ij} < \rho_{ij}, \quad A_0^s(i, j) = A_0(i, j) \mathbf{1}_{\{\varepsilon_{ij}^2 < s < \rho_{ij}^2\}}.$$

The chosen endpoints are fixed independently of the Gaussian centers. Keep A_1, A_2 fixed and form the weighted matrices and costs using A_0^s in place of A_0 . The arrays are complex; any phase at an output pair may be included in the fixed value $A_0(i, j)$. Set

$$s_- = \min_{i,j} \varepsilon_{ij}^2, \quad s_+ = \max_{i,j} \rho_{ij}^2.$$

Proposition 5.3 (Dissipation with output-dependent endpoints). *For every $h > 0$, $N \geq 0$, complex arrays A_v on $\{-N, \dots, N\}^2$, and endpoint choices as above, let α be any permutation of $(1, 11/10, 11/10)$ and use variances $\sigma_v = \alpha_v s$. The corresponding masked costs satisfy*

$$\int_{s-}^{s+} \int_{\Pi_0} \sum_{v=0}^2 V_v(p, s) \, d\pi \, ds \leq C\mathcal{S}(A). \quad (5.5)$$

The constant is independent of the arrays, the mesh, the endpoints, and the number of distinct endpoint values.

Proof. Write $J(s) = \int_{\Pi_0} \sum_v e_v(p, s) \, d\pi$ for the masked energy, and put $\Delta J(\tau) = J(\tau+) - J(\tau-)$ at an endpoint. We first estimate its jump at a single scale, then sum those estimates and telescope the heat inequality.

The two stars at a jump. At an endpoint scale τ , hold the weights at $s = \tau$ and let

$$\delta A(i, j) = A_0^{\tau+}(i, j) - A_0^{\tau-}(i, j).$$

The superscripts denote one-sided masks. Interpolate all changed entries at once between these two arrays. Each interpolated entry has absolute value at most $|A_0(i, j)|$, even when some entries open and others close at the same scale.

For fixed i, j, p , define nonnegative numbers using the original, unmasked arrays:

$$\begin{aligned} X^2 &= \sum_m |A_0(i, m)|^2 w_1(m), & Y^2 &= \sum_m |A_0(m, j)|^2 w_0(m), \\ H^2 &= \sum_m |A_2(m, i)|^2 w_2(m), & K^2 &= \sum_m |A_1(j, m)|^2 w_2(m). \end{aligned} \quad (5.6)$$

In $R_0 = [W_0 \, W_2^*]$, a changing entry has row norm at most $\sqrt{w_0(i)}\sqrt{X^2 + H^2}$, column norm at most $\sqrt{w_1(j)}Y$, and derivative modulus $\sqrt{w_0(i)w_1(j)}|\delta A(i, j)|$. In $R_1 = [W_1 \, W_0^*]$, its adjoint entry has row norm at most $\sqrt{w_1(j)}\sqrt{Y^2 + K^2}$ and column norm at most $\sqrt{w_0(i)}X$, with the same derivative modulus. Star $R_2 = [W_2 \, W_1^*]$ is unchanged. Lemma 5.2 and $\sqrt{x^2 + y^2} \leq x + y$ therefore give

$$|\Delta e_0| + |\Delta e_1| \leq 3 \sum_{i,j} |\delta A(i, j)| w_0(i) w_1(j) [(X + H)Y + (Y + K)X]. \quad (5.7)$$

This estimate requires no assumption on the ranks during interpolation.

A probability measure on the center plane. For each fixed i, j , the measure

$$d\nu_{ij} = h^{-2} w_0(i) w_1(j) \, d\pi$$

is a probability measure. Indeed, in the free coordinates (p_0, p_1) , its density is $g_{\sigma_0}(u_i - p_0)g_{\sigma_1}(u_j - p_1)$. Thus p_0, p_1 are independent Gaussian variables with means u_i, u_j and variances σ_0, σ_1 , while $p_2 = -p_0 - p_1$. Adding Gaussian variances gives

$$\begin{aligned} \mathbb{E}_{\nu_{ij}} w_0(m) &= h g_{2\sigma_0}(u_m - u_i), \\ \mathbb{E}_{\nu_{ij}} w_1(m) &= h g_{2\sigma_1}(u_m - u_j), \\ \mathbb{E}_{\nu_{ij}} w_2(m) &= h g_{\sigma_0 + \sigma_1 + \sigma_2}(u_m + u_i + u_j). \end{aligned} \quad (5.8)$$

For the last identity, $p_0 + p_1$ has mean $u_i + u_j$ and variance $\sigma_0 + \sigma_1$, and averaging $g_{\sigma_2}(u_m + p_0 + p_1)$ adds the remaining variance. Each identity also follows directly by completing the square in its Gaussian integral.

Extend the arrays by zero to $\mathbb{Z}^2$ and define

$$\begin{aligned} X_*^2(i, j) &= M_{\mathbb{Z}}(|A_0(i, \cdot)|^2)(j), \\ Y_*^2(i, j) &= M_{\mathbb{Z}}(|A_0(\cdot, j)|^2)(i), \\ H_*^2(i, j) &= M_{\mathbb{Z}}(|A_2(\cdot, i)|^2)(-i - j), \\ K_*^2(i, j) &= M_{\mathbb{Z}}(|A_1(j, \cdot)|^2)(-i - j). \end{aligned} \tag{5.9}$$

Every variance in (5.8) is at least h^2 . Lemma 5.1 consequently bounds $\mathbb{E}X^2, \mathbb{E}Y^2, \mathbb{E}H^2, \mathbb{E}K^2$ by C_g times the corresponding starred squares. Cauchy–Schwarz under the probability ν_{ij} gives, in particular,

$$\mathbb{E}(XY) \leq C_g X_* Y_*, \quad \mathbb{E}(HY) \leq C_g H_* Y_*, \quad \mathbb{E}(KX) \leq C_g K_* X_*.$$

No independence of these four random variables is needed. The starred quantities depend only on the original arrays and the output indices, not on the scale τ .

Summing the jumps. Integrating (5.7) over the centers and using $w_0(i)w_1(j) \, d\pi = h^2 \, d\nu_{ij}$ yields

$$|\Delta J(\tau)| \leq 3C_g \sum_{i,j} h^2 |\delta A(i, j)| (2X_* Y_* + H_* Y_* + K_* X_*).$$

Every entry opens once and closes once. Therefore its total absolute change, summed over any subset of endpoint scales, is at most $2|A_0(i, j)|$. We obtain

$$\sum_{\tau} |\Delta J(\tau)| \leq 6C_g \sum_{i,j} h^2 |A_0(i, j)| (2X_* Y_* + H_* Y_* + K_* X_*). \tag{5.10}$$

The sum may include all endpoints or just the interior ones.

To bound this expression, set

$$n_v = \left(\sum_{i,j} h^2 |A_v(i, j)|^3 \right)^{1/3}, \quad \|f\|_{3,h}^3 = \sum_{i,j \in \mathbb{Z}} h^2 |f(i, j)|^3.$$

The $q = 3/2$ case of (5.1), applied on slices, gives

$$\|X_*\|_{3,h}, \|Y_*\|_{3,h} \leq Cn_0, \quad \|H_*\|_{3,h} \leq Cn_2, \quad \|K_*\|_{3,h} \leq Cn_1. \tag{5.11}$$

For H_* , fix i and change the summation index from j to $-i - j$; this is a bijection of $\mathbb{Z}$. For K_* , fix j and change i to $-i - j$. Thus both estimates are ordinary slice estimates, with no multiplicity or mesh factor. Restricting the sums back to the original finite square can only decrease them.

Hölder's inequality in (5.10), followed by (5.11), proves

$$\sum_{\tau} |\Delta J(\tau)| \leq C(2n_0^3 + n_0^2 n_2 + n_0^2 n_1) \leq C\mathcal{S}(A). \tag{5.12}$$

The last step uses $a^2 b \leq (2a^3 + b^3)/3$. In particular, the estimate has no dependence on how many distinct scales occur.

Telescoping the heat estimate. List the distinct squared endpoints as $t_0 = s_- < t_1 < \dots < t_m = s_+$. On every open interval (t_k, t_{k+1}) the array is fixed, so Lemma 4.4 applies. Its fixed-array continuity and Gaussian domination also justify the one-sided endpoint limits. Integrating on these intervals and summing gives

$$\begin{aligned} c_* \int_{s_-}^{s_+} \int_{\Pi_0} \sum_v V_v \, d\pi \, ds &\leq \sum_{k=0}^{m-1} [J(t_k+) - J(t_{k+1}-)] \\ &= J(s_-+) - J(s_+-) + \sum_{k=1}^{m-1} [J(t_k+) - J(t_k-)] \\ &\leq J(s_-+) + \sum_{k=1}^{m-1} |\Delta J(t_k)|. \end{aligned}$$

Here the terminal energy was discarded by nonnegativity. The initial right-hand mask has entries bounded by those of the original array, and all its variances are at least h^2 . Lemma 4.5 therefore gives $J(s_-+) \leq CS(A)$. Together with (5.12), this proves (5.5). The initial opening is accounted for by the right-hand initial energy; a final closing jump is unnecessary. Simultaneous endpoints have already been combined in the list of distinct scales, and their values at the single endpoint do not affect the scale integral. $\square$

Figure 2 illustrates the masks and the energy accounting.

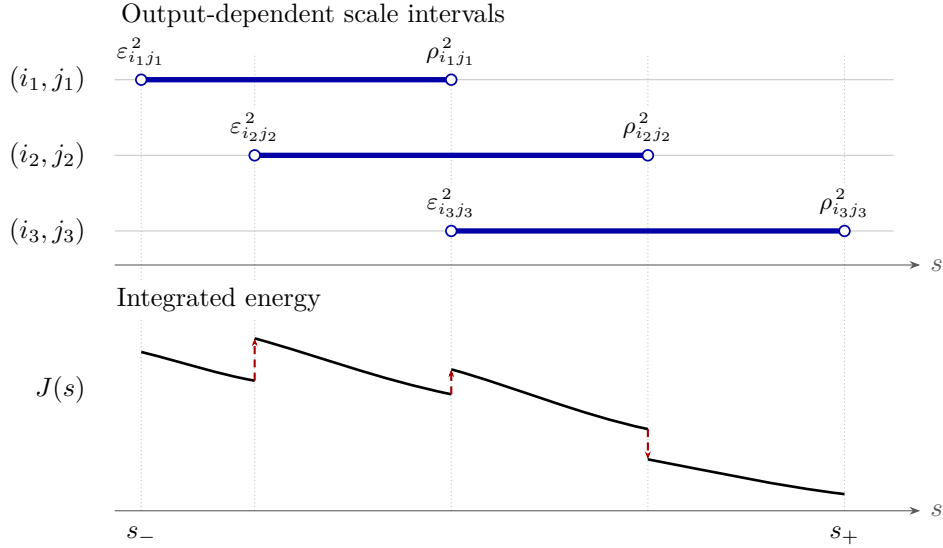


Figure 2: Three selected output pairs and a schematic integrated energy, not an exact heat-flow solution. The entry $A_0^s(i, j)$ is retained only on $(\varepsilon_{ij}^2, \rho_{ij}^2)$. Between successive event scales, $J(s)$ is nonincreasing by the fixed-array dissipation inequality. Dashed arrows connect its one-sided limits at events; the values at the events themselves are not shown. Jumps may have either sign, and endpoints may coincide, as illustrated by $\rho_{i_1 j_1}^2 = \varepsilon_{i_3 j_3}^2$. The bound $\sum_\tau |\Delta J(\tau)| \leq CS(A)$, together with the initial-energy bound, controls the accumulated dissipation independently of the number of event scales.

6 From masked energies to the maximal transform

The estimate of Section 5 allows different truncation endpoints at every output position. We first use this freedom on a finite mesh, choosing the endpoints that maximize each output.

We then pass to functions and compare the resulting smooth kernels with hard truncations. The comparison will be pointwise and uniform in both endpoints.

6.1 Odd windows and selected endpoints

For $s > 0$ and $0 < \varepsilon < R$, define

$$k_s(t) = \frac{t}{\sqrt{s}} g_s(t), \quad \psi = k_1 * k_1 * k_1, \quad I_{\varepsilon, R}(t) = \int_{\varepsilon^2}^{R^2} s^{-1/2} \psi(t/\sqrt{s}) \frac{ds}{s}.$$

Here $*$ is convolution on $\mathbb{R}$. The function ψ is Schwartz, and $I_{\varepsilon, R}$ is smooth and integrable for fixed positive endpoints.

Proposition 6.1 (Selected smooth truncations). *Let $h > 0$, let $N \geq 0$ be an integer, set $u_i = ih$, and let $A_v(i, j)$, $v = 0, 1, 2$, be complex arrays indexed by $-N \leq i, j \leq N$. Assign an arbitrary endpoint pair $h \leq \varepsilon_{ij} < \rho_{ij} < \infty$ to each (i, j) . Then*

$$\left| h^3 \sum_{i, j, l} A_0(i, j) A_1(j, l) A_2(l, i) I_{\varepsilon_{ij}, \rho_{ij}}(u_i + u_j + u_l) \right| \leq C \prod_{v=0}^2 \left(\sum_{i, j} h^2 |A_v(i, j)|^3 \right)^{1/3}. \quad (6.1)$$

The constant is independent of the arrays, mesh and selected endpoints.

Proof. We apply the mixed trace inequality with Gaussian weights of larger variance than the windows k_s , so that the diagonal insertions are bounded. We then compare each quadratic cost with a variance choice covered by Proposition 5.3.

Use the masked array $A_0^s(i, j) = A_0(i, j) \mathbf{1}_{\{\varepsilon_{ij}^2 < s < \rho_{ij}^2\}}$ and the scale interval $[s_-, s_+]$ from Proposition 5.3. All weighted matrices below use this same mask. Let superscript b denote the base variances $(\lambda s, \lambda s, \lambda s)$, where $\lambda = 11/10$, and put

$$D_v(i, i) = \frac{k_s(u_i - p_v)}{g_{\lambda s}(u_i - p_v)}.$$

These matrices are real diagonal. With $z = (u_i - p_v)/\sqrt{s}$,

$$|D_v(i, i)| = \sqrt{\lambda} |z| e^{-(\lambda-1)z^2/(2\lambda)} \leq \frac{\lambda}{\sqrt{e(\lambda-1)}}.$$

Thus their operator norms have an absolute bound.

For each vertex v , let $\alpha(v)$ equal 1 at v and λ at the other two vertices. In the star centered at v , changing from the base variances to $\alpha(v)s$ changes only its row weight. In particular,

$$R_v^{\alpha(v)} = \text{diag}_i \left(\sqrt{\frac{g_s(u_i - p_v)}{g_{\lambda s}(u_i - p_v)}} \right) R_v^b, \quad T_v^{\alpha(v)} \leq \sqrt{\lambda} T_v^b.$$

The last inequality uses $g_s \leq \sqrt{\lambda} g_{\lambda s}$. The score $L_v^{\alpha(v)}$ equals $\text{diag}_i((u_i - p_v)/s)$, so the row densities also give the exact identity

$$(R_v^b)^* D_v R_v^b = \sqrt{s} (R_v^{\alpha(v)})^* L_v^{\alpha(v)} R_v^{\alpha(v)}. \quad (6.2)$$

Indeed, both inserted row densities are $h k_s(u_i - p_v)$, and all column weights agree. The insertion on the left is therefore supported on $T_v^{\alpha(v)}$. Lemma 2.2 and quadratic homogeneity give

$$b_{T_v^b}((R_v^b)^* D_v R_v^b) \leq \lambda^{1/4} s V_v^{\alpha(v)}.$$

Apply Lemma 3.1 to the base matrices. Integrate its three nonnegative terms separately, using Proposition 5.3 with the corresponding $\alpha(v)$. The factor s cancels ds/s , and we obtain

$$\int_{s_-}^{s_+} \int_{\Pi_0} \left| \text{tr}(D_0 W_0^b D_1 W_1^b D_2 W_2^b) \right| d\pi_0 \frac{ds}{s} \leq C\mathcal{S}(A). \quad (6.3)$$

We identify the signed integral in this estimate. Expanding the trace leaves one signed Gaussian weight at each vertex:

$$h^3 \sum_{i,j,l} A_0^s(i,j) A_1(j,l) A_2(l,i) k_s(u_i - p_0) k_s(u_j - p_1) k_s(u_l - p_2).$$

On Π_0 , substitute $r = u_i - p_0$ and $q = u_j - p_1$. The Jacobian is one and the third argument becomes $u_i + u_j + u_l - r - q$. Consequently the center integral is $(k_s * k_s * k_s)(u_i + u_j + u_l)$. Since $k_s(t) = s^{-1/2} k_1(t/\sqrt{s})$, this convolution equals

$$s^{-1/2} \psi((u_i + u_j + u_l)/\sqrt{s}).$$

The sums are finite and the Gaussian products are absolutely integrable on the center plane and the compact positive scale interval. Thus all these integrations can be performed term by term. The mask restricts the scale integral of the (i,j,l) term to $(\varepsilon_{ij}^2, \rho_{ij}^2)$. Taking the absolute value after integration in (6.3) proves (6.1) with its right side initially replaced by $C\mathcal{S}(A)$.

Put $n_v = (\sum h^2 |A_v|^3)^{1/3}$. If any n_v is zero the signed sum vanishes. Otherwise apply the last estimate to A_v/n_v for all three vertices, keeping the selected endpoints fixed. The normalized value of $\mathcal{S}$ is 3. Scaling the signed trilinear expression back proves the proposition. $\square$

6.2 Finite maxima and the continuum limit

For functions F, G on $\mathbb{R}^2$, write

$$B_{\varepsilon,R}^I(F,G)(x,y) = \int_{\mathbb{R}} F(x+t,y) G(x,y+t) I_{\varepsilon,R}(t) dt.$$

For compactly supported smooth inputs this integral is defined at every point. We now place the supremum over its endpoints inside the spatial norm.

Proposition 6.2. *For all complex $F, G \in C_c^\infty(\mathbb{R}^2)$,*

$$\left\| \sup_{0 < \varepsilon < R} |B_{\varepsilon,R}^I(F,G)| \right\|_{L^{3/2}(\mathbb{R}^2)} \leq C \|F\|_{L^3(\mathbb{R}^2)} \|G\|_{L^3(\mathbb{R}^2)}.$$

The supremum is measurable and the constant is absolute.

Proof. First fix a nonempty finite family of endpoint pairs (ε_k, R_k) , $1 \leq k \leq m$, and take $h \leq \min_k \varepsilon_k$. For finite arrays A_1, A_2 , define

$$U_k(i,j) = h \sum_l A_1(j,l) A_2(l,i) I_{\varepsilon_k, R_k}(u_i + u_j + u_l), \quad M(i,j) = \max_{1 \leq k \leq m} |U_k(i,j)|.$$

At each output pair choose the least index attaining this maximum, and write $U(i,j)$ for its output. These choices provide admissible $\varepsilon_{ij}, \rho_{ij}$ in Proposition 6.1. They are fixed while the testing array A_0 varies.

Set $H = \sum_{i,j} h^2 M(i,j)^{3/2}$. If $H = 0$ the desired bound is immediate. Otherwise choose

$$A_0(i,j) = \begin{cases} H^{-1/3} M(i,j)^{1/2} \overline{U(i,j)} / |U(i,j)|, & U(i,j) \neq 0, \\ 0, & U(i,j) = 0. \end{cases}$$

Then $\sum h^2 |A_0|^3 = 1$ and $h^2 \sum A_0 U = H^{2/3}$. Proposition 6.1 therefore gives

$$\left(\sum_{i,j} h^2 M(i,j)^{3/2} \right)^{2/3} \leq C \prod_{v=1}^2 \left(\sum_{i,j} h^2 |A_v(i,j)|^3 \right)^{1/3}. \quad (6.4)$$

This estimate has no dependence on the number of endpoint pairs. The complex phase of A_0 accounts for the absolute value at each output.

To pass to functions, define the compactly supported smooth inputs

$$G_1(b,c) = F(b+c, -b), \quad G_2(c,a) = G(-a, c+a).$$

Both pair maps have determinant one, so $\|G_1\|_3 = \|F\|_3$ and $\|G_2\|_3 = \|G\|_3$. Choose $B > 0$ so that both supports lie in the interior of $[-B, B]^2$. Take $h = B/N$, $-N \leq i, j, l \leq N$, and sample

$$A_1(j,l) = G_1(jh, lh), \quad A_2(l,i) = G_2(lh, ih).$$

The corresponding continuum outputs are

$$\mathcal{U}_k(a,b) = \int_{\mathbb{R}} G_1(b,c) G_2(c,a) I_{\varepsilon_k, R_k}(a+b+c) \, dc.$$

They vanish outside $[-B, B]^2$: $G_2(c,a)$ restricts a and $G_1(b,c)$ restricts b . A nonzero integrand also has $c \in [-B, B]$. For our fixed finite endpoint family the integrands and their c derivatives are uniformly bounded on this compact cube. Riemann sums in c consequently converge uniformly for $(a,b) \in [-B, B]^2$ and $1 \leq k \leq m$. In particular,

$$\max_{i,j} \max_{1 \leq k \leq m} |U_k(i,j) - \mathcal{U}_k(ih, jh)| \longrightarrow 0.$$

The finite maximum of magnitudes preserves this uniform convergence. Since $\max_k |\mathcal{U}_k|$ is continuous and compactly supported, the two-dimensional Riemann sums on the left of (6.4) converge to its $L^{3/2}$ norm. The cube roots of the input sums converge to $\|G_1\|_3$ and $\|G_2\|_3$. The condition on h holds for all sufficiently large N . We obtain

$$\| \max_k |\mathcal{U}_k| \|_{3/2} \leq C \|G_1\|_3 \|G_2\|_3.$$

No convergence or regularity of the maximizing indices is needed: we passed to the limit in the finite maximum itself.

With $x = -a$, $y = -b$, and $t = a + b + c$, the last integral is $B_{\varepsilon_k, R_k}^I(F, G)(x, y)$. The output coordinate change preserves Lebesgue measure, giving the asserted estimate for every finite endpoint family.

Finally enumerate all rational pairs $0 < \varepsilon < R$ and take increasing finite subfamilies. At each spatial point the output is continuous in the two positive endpoints, by its defining integrals on compact scale and spatial intervals. Thus the supremum over these rational pairs equals the supremum over all real pairs. It is measurable as a countable supremum of continuous spatial functions. Monotone convergence of the $3/2$ powers of the finite maxima proves the result. $\square$

6.3 A pointwise comparison with hard truncations

The smooth maximal estimate will imply the hard one because each endpoint error has a common pointwise majorant. A uniform bound on its kernel's L^1 norm alone would not provide this comparison.

Lemma 6.3. *There are an absolute constant C and functions E_r , $r > 0$, such that, for $t \neq 0$, apart from endpoint values,*

$$I_{\varepsilon,R}(t) = \frac{c_0}{t} \mathbf{1}_{\{\varepsilon < |t| < R\}} + E_\varepsilon(t) - E_R(t), \quad c_0 = -\frac{2}{3\sqrt{6\pi}}, \quad (6.5)$$

and

$$|E_r(t)| \leq \frac{C}{r} \left(1 + \frac{|t|}{r}\right)^{-2} \quad (t \neq 0). \quad (6.6)$$

Proof. Since $k_1 = -g_1'$ and Gaussian variances add under convolution,

$$\psi(t) = -g_3'''(t) = \left(\frac{t^3}{27} - \frac{t}{3}\right) g_3(t).$$

The convolution identity follows by completing the square; differentiation under convolution is justified by Schwartz decay. For $v \geq 0$, put

$$P(v) = 2 \int_0^v \psi(z) \, dz = 2g_3''(0) - 2g_3''(v), \quad c_0 = P(\infty) = 2g_3''(0) = -\frac{2}{3\sqrt{6\pi}}.$$

The substitution $z = t/\sqrt{s}$ for $t > 0$, followed by oddness for $t < 0$, yields

$$I_{\varepsilon,R}(t) = \frac{P(|t|/\varepsilon) - P(|t|/R)}{t}.$$

Define $Q(v) = P(v) - c_0 \mathbf{1}_{\{v > 1\}}$ for $v \geq 0$. Smooth oddness of ψ gives $P(v) = O(v^2)$ at zero. For $v > 1$,

$$Q(v) = -2g_3''(v) = -2 \left(\frac{v^2}{9} - \frac{1}{3}\right) g_3(v).$$

Thus Q has Gaussian decay at infinity and bounded one-sided values at its sole jump, $v = 1$. Enlarging an absolute constant on a compact interval, we have

$$\frac{|Q(v)|}{v} \leq C(1+v)^{-2} \quad (v > 0).$$

Set $E_r(t) = Q(|t|/r)/t$ for $t \neq 0$ and set $E_r(0) = 0$. Substituting the definition of Q proves (6.5) and (6.6). Values at the four hard endpoints have no effect on any integral. $\square$

Let M_x and M_y denote the continuous centered interval maximal operators in the first and second variables, respectively. The next estimate follows from their one-dimensional bounds in Lemma 5.1.

Lemma 6.4. *For complex $F, G \in C_c^\infty(\mathbb{R}^2)$ and every $r > 0$,*

$$\left| \int_{\mathbb{R}} F(x+t, y) G(x, y+t) E_r(t) \, dt \right| \leq C [M_x(|F|^2)(x, y) M_y(|G|^2)(x, y)]^{1/2}.$$

The right side has $L^{3/2}(\mathbb{R}^2)$ norm at most $C\|F\|_3\|G\|_3$.

Proof. Put $q_r(t) = r^{-1}(1 + |t|/r)^{-2}$. For nonnegative locally integrable f on $\mathbb{R}$, the part of $\int q_r(t)f(x+t) dt$ with $|t| \leq r$ is at most $2M_{\mathbb{R}}f(x)$. On the shell $2^k r < |t| \leq 2^{k+1}r$, $k \geq 0$, its contribution is at most

$$r^{-1}2^{-2k} \int_{|t| \leq 2^{k+1}r} f(x+t) dt \leq 4 \cdot 2^{-k} M_{\mathbb{R}}f(x).$$

Summing the geometric series proves a bound independent of r . By (6.6) and weighted Cauchy–Schwarz,

$$\left| \int F(x+t, y)G(x, y+t)E_r(t) dt \right| \leq C \left(\int |F(x+t, y)|^2 q_r(t) dt \right)^{1/2} \cdot \left(\int |G(x, y+t)|^2 q_r(t) dt \right)^{1/2}.$$

Apply the preceding one-dimensional bound on the respective coordinate lines. This gives the pointwise assertion.

Hölder’s inequality and Lemma 5.1, applied on slices and then integrated in the other coordinate, give

$$\begin{aligned} \|[M_x(|F|^2)M_y(|G|^2)]^{1/2}\|_{3/2} &\leq \|M_x(|F|^2)\|_{3/2}^{1/2} \|M_y(|G|^2)\|_{3/2}^{1/2} \\ &\leq C \| |F|^2 \|_{3/2}^{1/2} \| |G|^2 \|_{3/2}^{1/2} = C \|F\|_3 \|G\|_3. \end{aligned}$$

□

Proposition 6.5 (Compact smooth inputs). *For all complex $F, G \in C_c^\infty(\mathbb{R}^2)$,*

$$\|B_*(F, G)\|_{L^{3/2}(\mathbb{R}^2)} \leq C \|F\|_{L^3(\mathbb{R}^2)} \|G\|_{L^3(\mathbb{R}^2)},$$

with an absolute constant. The supremum defining B_ is over both finite hard truncation endpoints, inside the spatial norm.*

Proof. Lemma 6.3 and Lemma 6.4 give, at each spatial point,

$$|c_0| \sup_{0 < \varepsilon < R} |B_{\varepsilon, R}(F, G)| \leq \sup_{0 < \varepsilon < R} |B_{\varepsilon, R}^I(F, G)| + 2C [M_x(|F|^2)M_y(|G|^2)]^{1/2}.$$

Both endpoint errors have the same majorant, independently of their radii. The hard outputs are continuous in both positive endpoints, so their supremum also equals the measurable supremum over rational pairs. Take $L^{3/2}$ norms, apply Proposition 6.2 and Lemma 6.4, and divide by $|c_0| > 0$. □

7 General inputs and principal values

We now pass the hard maximal estimate from compactly supported smooth inputs to all complex L^3 inputs. The main point is to control all truncations on one set of full measure. We then prove joint principal-value convergence and recover the scalar simplex formulation.

Write $p = 3/2$ and

$$B_{\varepsilon, R}(F, G)(x, y) = \int_{\varepsilon < |t| < R} F(x+t, y)G(x, y+t) \frac{dt}{t}.$$

Proposition 6.5 supplies the hard maximal estimate for compactly supported smooth inputs; denote its constant by C_M .

Lemma 7.1 (Finite truncations). *For complex $F, G \in L^3(\mathbb{R}^2)$, there is a set of full measure on which every finite truncation is absolutely convergent and depends continuously on (ε, R) in $0 < \varepsilon < R < \infty$. The resulting family is independent of the representatives of F, G up to a common null set. Moreover,*

$$\left\| \int_{\varepsilon < |t| < R} |F(x+t, y)G(x, y+t)| \frac{dt}{|t|} \right\|_p \leq 2 \log(R/\varepsilon) \|F\|_3 \|G\|_3. \quad (7.1)$$

Proof. For each fixed t , Hölder's inequality and translation invariance give $\|F(\cdot + t, \cdot)G(\cdot, \cdot + t)\|_p \leq \|F\|_3 \|G\|_3$. Minkowski's integral inequality proves (7.1). Apply it to the countable annuli $1/n < |t| < n$, $n \geq 2$. Outside their combined exceptional null set, the integrand is absolutely integrable on every compact annulus away from zero. Absolute continuity of its integral gives the asserted endpoint continuity. If either input changes on a null set, apply the same bound to the zero L^3 difference, and use the same countable annuli. This proves independence simultaneously at every endpoint pair. $\square$

Define $B_*(F, G)$ on this full-measure set as the supremum of $|B_{\varepsilon, R}(F, G)|$ over all finite endpoint pairs, and set it to zero elsewhere. Endpoint continuity makes this supremum equal to the one over positive rational endpoint pairs, so it is measurable.

Proof of Theorem 1.1. Choose compactly supported smooth $f_m \rightarrow F$ and $g_m \rightarrow G$ in L^3 . For a fixed endpoint pair, bilinearity and (7.1) show that $B_{\varepsilon, R}(f_m, g_m) \rightarrow B_{\varepsilon, R}(F, G)$ in L^p . Fix a finite family of rational endpoint pairs. The difference of the two finite maxima of magnitudes is bounded by the sum of the corresponding truncation differences. Thus these maxima also converge in L^p . Passing the compact smooth maximal estimate to the limit bounds this finite maximum by $C_M \|F\|_3 \|G\|_3$. The constant is independent of the family. Exhaust all rational endpoint pairs and use monotone convergence and Lemma 7.1 to obtain Theorem 1.1.

We also obtain a useful continuity statement. For two pairs of inputs, bilinearity of every truncation gives, almost everywhere,

$$|B_*(F, G) - B_*(f, g)| \leq B_*(F - f, G) + B_*(f, G - g). \quad (7.2)$$

The bound just established makes the maximal map continuous from $L^3 \times L^3$ to $L^{3/2}$. Density shows that it is the unique continuous extension of its compact smooth values. $\square$

Theorem 7.2 (Joint bilinear principal values). *For all complex $F, G \in L^3(\mathbb{R}^2)$, the joint limit*

$$B(F, G) = \lim_{\substack{\varepsilon \downarrow 0 \\ R \uparrow \infty}} B_{\varepsilon, R}(F, G)$$

exists almost everywhere and in $L^{3/2}(\mathbb{R}^2)$, with $\|B(F, G)\|_{3/2} \leq C_M \|F\|_3 \|G\|_3$. More precisely,

$$\left\| \sup_{\substack{0 < \varepsilon < 1/n \\ R > n}} |B_{\varepsilon, R}(F, G) - B(F, G)| \right\|_{3/2} \longrightarrow 0. \quad (7.3)$$

The limit is the unique bounded bilinear extension of the principal-value operator on compactly supported smooth inputs.

Proof. For compactly supported smooth f, g and a fixed (x, y) , the function $h(t) = f(x+t, y)g(x, y+t)$ is smooth and compactly supported. The truncated integral is

$$\int_{\varepsilon}^R \frac{h(t) - h(-t)}{t} dt.$$

Its numerator is $O(t)$ at zero and vanishes for large t , so it has a joint limit at every point.

For general inputs, let

$$\Omega_n(F, G) = \sup_{\substack{0 < \varepsilon, \varepsilon' < 1/n \\ R, R' > n}} |B_{\varepsilon, R}(F, G) - B_{\varepsilon', R'}(F, G)|.$$

These are measurable: by endpoint continuity the supremum may be taken over rational parameters. They decrease with n and are bounded by $2B_*(F, G)$. Fix smooth approximants f_m, g_m as above. Outside a single null set for this countable collection of input pairs,

$$\Omega_n(F, G) \leq \Omega_n(f_m, g_m) + 2B_*(F - f_m, G) + 2B_*(f_m, G - g_m).$$

Letting $n \rightarrow \infty$, the smooth joint convergence and Theorem 1.1 give

$$\left\| \lim_{n \rightarrow \infty} \Omega_n(F, G) \right\|_p \leq 2C_M (\|F - f_m\|_3 \|G\|_3 + \|f_m\|_3 \|G - g_m\|_3).$$

The right side tends to zero with m . Hence the limiting oscillation is zero almost everywhere. This is the Cauchy criterion for all pairs of endpoints in the indicated tails, so the joint complex limit exists. It is measurable by taking a fixed cofinal sequence of endpoint pairs, and its magnitude is at most $B_*(F, G)$.

Taking (ε', R') to the limit in the definition of Ω_n bounds the supremum in (7.3) by $\Omega_n(F, G)$. Dominated convergence applies because $0 \leq \Omega_n(F, G) \leq 2B_*(F, G) \in L^p$. This proves (7.3), and hence joint norm convergence. Passing to the norm limit in the bilinear identities for the truncations proves bilinearity of B . Its bound and uniqueness follow from Theorem 1.1 and density. $\square$

7.1 Scalar forms and the simplex coordinates

For complex $F_0, F_1, F_2 \in L^3(\mathbb{R}^2)$, define the flat scalar form

$$\mathcal{L}_{\varepsilon, R}(F_0, F_1, F_2) = \iiint_{\varepsilon < |t| < R} F_0(x, y) F_1(x + t, y) F_2(x, y + t) \, dx \, dy \, \frac{dt}{t}.$$

Corollary 7.3 (Flat scalar principal values). *For complex $F_0, F_1, F_2 \in L^3(\mathbb{R}^2)$, every finite flat truncation is absolutely integrable, and*

$$\sup_{0 < \varepsilon < R < \infty} |\mathcal{L}_{\varepsilon, R}(F_0, F_1, F_2)| \leq C_M \prod_{v=0}^2 \|F_v\|_3.$$

Its joint scalar principal value exists and equals $\int F_0 B(F_1, F_2)$. This limiting form is the unique bounded trilinear extension of its compact smooth definition.

Proof. Hölder's inequality in (x, y) bounds the absolute integral by $2 \log(R/\varepsilon) \prod_v \|F_v\|_3$. Thus Fubini identifies the truncated form with $\int F_0 B_{\varepsilon, R}(F_1, F_2)$. Theorem 1.1 gives its endpoint-uniform bound, and Theorem 7.2 gives joint scalar convergence by pairing the $L^{3/2}$ limit with F_0 . The limiting form is bounded and complex trilinear. Density of compact smooth triples proves uniqueness. $\square$

For the simplex form, set

$$\Lambda_{\varepsilon, R}(G_0, G_1, G_2) = \iiint_{\varepsilon < |a+b+c| < R} G_0(a, b) G_1(b, c) G_2(c, a) \frac{da \, db \, dc}{a + b + c}.$$

The flat and simplex scalar formulations, including equality of their optimal norm constants, are already related in [13, Appendix B.1]. We record explicit norm-preserving maps and track both finite endpoints and their joint limit.

Proposition 7.4 (Scalar equivalence). *For all complex $G_0, G_1, G_2 \in L^3(\mathbb{R}^2)$, every finite simplex truncation is absolutely integrable, and*

$$\sup_{0 < \varepsilon < R < \infty} |\Lambda_{\varepsilon, R}(G_0, G_1, G_2)| \leq C_M \prod_{v=0}^2 \|G_v\|_3.$$

Its joint scalar principal value exists for every such triple and is the unique bounded trilinear extension of its compact smooth definition. The optimal constants in the uniform flat and simplex scalar estimates are equal.

Proof. Use the determinant-one change of variables

$$x = -a, \quad y = -b, \quad t = a + b + c, \quad a = -x, \quad b = -y, \quad c = t + x + y.$$

For flat inputs define

$$\begin{aligned} G_0(u, v) &= F_0(-u, -v), \\ G_1(u, v) &= F_1(u + v, -u), \\ G_2(u, v) &= F_2(-v, u + v). \end{aligned}$$

Each pair map has determinant one. Their inverses are

$$F_0(r, s) = G_0(-r, -s), \quad F_1(r, s) = G_1(-s, r + s), \quad F_2(r, s) = G_2(r + s, -r).$$

They preserve L^3 norms, compact smoothness, and Schwartz class. Direct substitution identifies the flat and simplex integrands and their truncation regions. The absolute integrability, uniform bound, and joint scalar convergence therefore follow from the flat conclusions. The inverse maps realize every simplex triple, which proves equality of the two optimal scalar constants. Density gives uniqueness of the bounded trilinear extension. $\square$

Write C_S for the common optimal scalar constant. Complex L^p duality shows that it is also the optimal constant for the uniform family of individual bounds

$$\|B_{\varepsilon, R}(F, G)\|_{3/2} \leq C_S \|F\|_3 \|G\|_3.$$

Indeed, for nonzero $U \in L^p$, the function

$$H = \frac{\bar{U}}{|U|} \frac{|U|^{1/2}}{\|U\|_p^{1/2}},$$

set to zero where $U = 0$, has L^3 norm one and satisfies $\int HU = \|U\|_p$. Apply this to each bilinear truncation; the zero case is immediate. The pointwise maximal optimal constant is at least C_S .

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